THE DE RHAM COHOMOLOGY OF A PROJECTIVE BUNDLE IS FREE OVER THE BASE, GENERATED BY POWERS OF THE FIRST CHERN CLASS OF THE TWISTING SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file derham.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-derham/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Projective space and its basic properties. Same chapter derham.tex, Section \ref{section-projective-space} (invoked at proof line 2268). It identifies the restriction of the projective bundle over a trivializing affine open with $\mathbf{P}^{r-1}_A$ and gives properness and smoothness of the projection.
- The de Rham cohomology of projective space. Same chapter derham.tex, label lemma-de-rham-cohomology-projective-space (invoked at proof line 2272). It shows that the powers $c_1^{dR}(\mathcal{O}_P(1))^i$, $i = 0, \ldots, r-1$, freely generate the de Rham cohomology of each fibre over its residue field.
- The criterion producing freeness over the base from a basis on every fibre. Same chapter derham.tex, label proposition-global-generation-on-fibres (invoked at proof line 2273). Applied to the proper smooth projection with the classes above, it turns the fibrewise basis statement into the asserted isomorphism.

Source reference. Original lemma 0FMJ: sources/stacks-derham/pinned-original-source.tex, source lines 1802–1802. Navigation pointer for \ref{lemma-de-rham-cohomology-projective-space} used inside the counted span (line 2272); the argument that consumes it is retained in the span.

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Source reference. Original proposition 0FMR: sources/stacks-derham/pinned-original-source.tex, source lines 2158–2158. Navigation pointer for \ref{proposition-global-generation-on-fibres} used inside the counted span (line 2273); the argument that consumes it is retained in the span.

Source reference. Original section 0FMG: sources/stacks-derham/pinned-original-source.tex, source lines 1576–1576. Navigation pointer for \ref{section-projective-space} used inside the counted span (line 2268); the argument that consumes it is retained in the span.

Proposition 0.1. *Let $X \to S$ be a morphism of schemes. Let $\mathcal{E}$ be a locally free $\mathcal{O}_X$-module of constant rank r. Consider the morphism $p : P = \mathbf{P}(\mathcal{E}) \to X$. Then the map*

$$\bigoplus\nolimits_{i=0,\ldots,r-1} H^*_{dR}(X/S) \longrightarrow H^*_{dR}(P/S)$$

given by the rule

$$(a_0, \ldots, a_{r-1}) \longmapsto \sum\nolimits_{i=0,\ldots,r-1} c_1^{dR}(\mathcal{O}_P(1))^i \cup p^*(a_i)$$

is an isomorphism.

Proof. Choose an affine open $\mathrm{Spec}(A) \subset X$ such that $\mathcal{E}$ restricts to the trivial locally free module $\mathcal{O}_{\mathrm{Spec}(A)}^{\oplus r}$. Then $P \times_X \mathrm{Spec}(A) = \mathbf{P}^{r-1}_A$. Thus we see that p is proper and smooth, see Section 0FMG. Moreover, the classes $c_1^{dR}(\mathcal{O}_P(1))^i$, $i = 0, 1, \ldots, r-1$ restricted to a fibre $X_y = \mathbf{P}^{r-1}_y$ freely generate the de Rham cohomology $H^*_{dR}(X_y/y)$ over $\kappa(y)$, see Lemma 0FMJ. Thus we've verified the conditions of Proposition 0FMR and we win. □